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ABSTRACT

These rapid advancements in deep fake technology have made it easier and more cost-effective to create highly realistic artificial videos and images. While such technologies can offer benefits in fields such as entertainment, media production, and digital communication, they also raise serious concerns regarding the security, privacy, and authenticity of digital media. The ability to manipulate visual content so convincingly has created significant challenges for verifying the credibility of online information. As a result, the detection of such manipulations has become an important research area. However, existing forensic detection techniques often struggle to generalize effectively across different datasets and various deepfake generation methods, limiting their reliability in real-world applications.

In this report, we propose a deep learning-based approach for effective deepfake video detection. To capture the temporal dependencies between successive video frames, the strong representational power of ResNeXt for spatial feature extraction is combined with Long Short-Term Memory (LSTM) networks. This hybrid architecture allows the model to analyze both spatial and temporal characteristics of video data simultaneously. By integrating these two components, the proposed method improves the ability to detect subtle inconsistencies such as minor facial expression changes, unnatural motion patterns, and temporal irregularities that are often overlooked by standalone frame-level detection methods. Furthermore, this study highlights the importance of hybrid deep learning architectures in addressing practical challenges in modern digital media forensics. The proposed approach contributes to the growing body of research aimed at enhancing the robustness, reliability, and realworld applicability of deepfake detection systems.

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LIST OF ABBREVIATIONS

LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
DF	Deep Forensics
AI	Artificial Intelligence
RNN	Recurrent Neural Network
ResNeXt	Aggregated Residual Transformations Network
GAN	Generative Adversarial Network
FF+	Face Forensics++
IoT	Internet of Things
Celeb-DF	Celebrity DeepFake Dataset
AWS	Amazon Web Services
SGD	Stochastic Gradient Descent
F1-Score	Harmonic Mean of Precision and Recall
DFDC	DeepFake Detection Challenge

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The term "deepfake", which typically describes how videos, images or audio of a person are digitally manipulated to create highly realistic but entirely false content, is a portmanteau of "deep learning" and "fake". Since first gaining prominence in 2017, deepfake technology has continually improved thanks to advances in artificial intelligence and generative deep learning technologies. New tools that use generative adversarial networks (GANs) and other deep neural network structures now allow anyone to produce realistic manipulated video and images with user-friendly software platforms. This has led to a lowered threshold for creating this type of media, and a consequent spread of manipulated media through social media, online journalism sites, and other digital platforms of communication.

Deepfakes present significant social and technological threats, such as misinformation, identity theft, political manipulation, fraud and privacy violations. Deepfake technology can be used to impersonate people, misdirect public opinion, or create evidence, thus causing digital media distrust. Moreover, the growing sophistication of deepfake videos makes it difficult for humans and machines to detect and identify manipulated videos. However, deepfake technology can also be used in a positive way for entertainment, filmmaking, accessibility, education and virtual communication. But the potential for abuse of such technology calls for an effective and efficient deepfake detection system. As a result, the detection and verification of manipulated videos as fake or genuine has emerged as a pressing research issue in the areas of cybersecurity and digital forensics. Although distinguishing between these fake films and authentic ones is proving to be an increasingly challenging task as they become more sophisticated, researchers are working on novel approaches to detect video manipulations [1] - [4].

To address these issues, advanced deepfake detection techniques have emerged in recent years. Conventional detection methods relying on hand-crafted features and classical machine learning methods are often limited in their ability to adapt to the ever-changing generation methods used to create deepfakes. These approaches generally rely on a few visual cues or inconsistencies, which current deepfake generation techniques increasingly avoid or remove. These traditional methods, therefore, often lack robustness across different datasets and manipulation types, limiting their practical applicability.

In recent years, researchers have increasingly explored deep learning approaches for fake video detection, which can learn discriminative features from a large corpus of video samples. These approaches exploit spatial and temporal cues in video frames to detect subtle anomalies in deepfake videos. Some research has proposed using convolutional neural

networks (CNNs) for extracting spatial features, while other work has integrated temporal modeling using Long Short-Term Memory (LSTM) networks to capture the temporal dynamics between video frames. Combined architectures, such as CNNs with recurrent neural networks (RNNs), such as CNN-LSTM models, have been successful in capturing spatiotemporal features of videos.

Moreover, ensemble models using different architectures like 2D and 3D CNNs have been developed to enhance detection accuracy by taking into account both spatial and temporal features. Others have explored frequency-based analysis, adversarial training, and other hybrid approaches to enhance the detection accuracy in difficult settings. Recently, new architectures, such as CNN-LSTM-Transformer models have been investigated to further improve the model's capability to detect manipulation in video content. These advances point to the arms race between more advanced methods of deepfake generation and forensic processes that detect deepfake videos [5]-[8].



Fig 1. Deep Fake Detection System

1.2 PROJECT DESCRIPTION

The exponential rise of deepfake technology poses a major threat to the authenticity and integrity of media content. Deepfake videos are produced through sophisticated artificial intelligence methods that can alter facial motions, speech and body movements to produce realistic but fictitious videos. These videos can be used to disinform, defame, and compromise security and privacy in various domains such as politics, finance, and social media. As deepfake videos become more realistic, traditional methods of detecting them which rely on manual feature extraction or simple machine learning methods are ineffective. Our system employs a novel deep learning framework that combines spatial and temporal modeling. This approach leverages the feature extraction power of the ResNeXt-50 convolutional neural network (CNN) to learn discriminative features and the temporal modeling capability of Long Short-Term Memory (LSTM) networks to capture temporal relationships in video. The ResNeXt-50 network extracts fine-grained spatial features from

each frame of a video sequence, focusing on cues related to lighting variations, texture inconsistencies, and anomalies along facial contours that may be present due to manipulation. The LSTM network considers temporal patterns between frames to detect inconsistencies in motion that could be indicative of deepfake manipulation.

For effective training and testing, the project uses standard benchmark datasets like Celeb-DF v1 and FaceForensics++. Using different datasets ensures that the system does not learn specific features unique to the datasets, which can help enhance real-world performance. Training the model on these datasets enables the system to identify deepfake videos in a variety of environments, lighting settings and manipulation techniques.

To train the model, the video frames undergo a range of preprocessing steps before being fed into the model. The frames are then normalised and resized to a consistent resolution for efficient training and consistent input dimensions. Facial alignment and detection techniques are also used to narrow the attention of the model to the facial area where deepfake manipulations typically occur. This helps remove distractions and accentuate features of interest, enabling the model to focus on the visual cues.

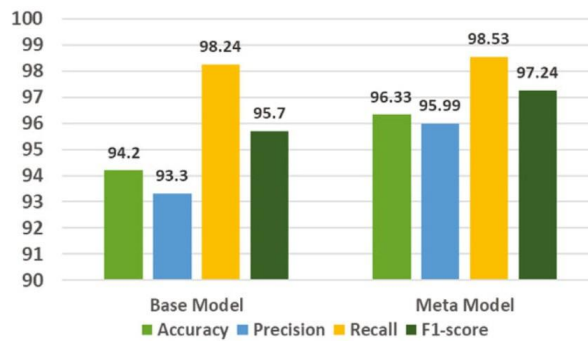


Fig 2. Performance comparison of the best base model and meta-learner tested on (a) Celeb-DF dataset

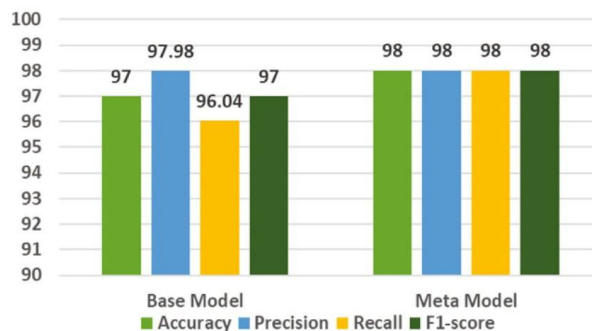


Fig 3. Performance comparison of the best base model and meta-learner tested on (a) FF+ dataset

In the training stage, the model is trained using supervised learning with equal numbers of authentic and manipulated video samples. Training includes multiple epochs and mini-batches to ensure learning is steady and overfitting is avoided. We use standard performance metrics such as accuracy, precision, recall, and F1-score to assess the model's performance. These measures offer insight into the effectiveness of the model in accurately detecting manipulated video while avoiding false alarms. The model is further validated through methods like cross-validation to assess its robustness on different parts of the dataset.

The experimental results indicate that the proposed hybrid model is effective in detecting deepfake videos. The system incorporates both spatial feature extraction and temporal sequence analysis, enabling it to identify subtle anomalies in facial expressions and motion patterns that are frequently overlooked by frame-level detection models. The results of the evaluation indicate an overall accuracy of around 97% as well as high precision and recall. This means that the model is able to distinguish real content from manipulated content. These results confirm the effectiveness of the combination of convolutional neural networks and recurrent networks for the deepfake detection task.

The project as a whole shows the potential of hybrid deep learning architectures to address the growing challenge of detecting deepfake media. The proposed framework exploits the spatial and temporal information available in videos to provide a more comprehensive and reliable detection scheme compared to conventional approaches. The system can be a useful tool for digital media forensics, cyber security applications and online content verification. While the computational requirements may preclude real-time deployment in some environments, the proposed approach provides the starting point for further research into the creation of more efficient and scalable deepfake detection systems.

CHAPTER 2

LITERATURE REVIEW

The availability and quality of datasets have been a key factor in shaping deepfake detection research. Several benchmark datasets have been constructed to facilitate training and evaluation of deep learning models for media manipulation detection. FaceForensics++ is one of the most popular datasets and has been widely used in experiments with convolutional neural network (CNN) based classifiers. Although this dataset allows for a structured environment for model building, it was generated under controlled conditions, which limits its ability to fully represent the complexities of real-world video manipulation scenarios. To overcome this limitation, the Celeb-DF dataset was introduced, which provides more diversity by incorporating videos of celebrities with different facial expressions, lighting conditions and environmental settings. This is an improvement but the dataset is still relatively small compared to the scale needed for robust model generalization. In addition, Facebook launched the DeepFake Detection Challenge (DFDC) in 2019 to overcome these limitations by providing a larger and more diverse dataset for research. However, even the best performing models showed limited success on the hidden test set, indicating the persistent challenge of developing detection models that generalize well to unseen data and new manipulation techniques [9]–[12].

Another critical challenge in deepfake detection research is the model generalization issue. Many detection models learn dataset-specific biases or noise rather than robust forensic cues that generalize across different datasets and manipulation techniques. Therefore, models with good results on one dataset might not perform well on another dataset created by different deepfake methods. This limitation is well-documented in the literature; several studies have shown that models trained on specific datasets do not always preserve accuracy when applied to real-world scenarios. This problem is further exacerbated by the fast pace of deepfake generation techniques which are constantly evolving new models that better conceal the visual artifacts and better mimic the natural facial movements.

Deepfake detection systems need to use modeling more. A lot of approaches looked at individual video frames. They did not look at how the frames related to each other. Deepfake videos often have small mistakes in how things move how faces look and how lips move. These mistakes can only be found when we look at the video over time. Some studies have shown that lip movements, strange head movements and faces that do not look right can mean the video has been manipulated. These things can help us find out if a video is a deepfake. Even though we know these things temporal modeling has not been used much in detection systems. So there is a lot of room for improvement, in deepfake detection systems. Temporal modeling can help us make deepfake detection systems better.

Researchers are looking into ways to detect things that are not real. They are trying to find things that're not normal in the way peoples bodies work, like how often they blink, their

heart rate or the small movements they make with their face. These things can be helpful in figuring out if a video or picture has been fake.. There are some problems with using these methods. Computers are getting better at making these things look real. It is harder to tell what is real and what is not. Also many of these methods need videos that're very clear which is a problem because a lot of the videos, on social media are not very clear.

Another limitation of deepfake detection is the computational cost associated with deep learning models. Some sophisticated detection models rely on deep neural network architectures which demand significant processing resources, as well as massive datasets, to train. This could restrict the actual implementation of such models in real-time security scenarios or large-scale monitoring systems. For instance, in an application which requires real-time detection of deepfakes within a video stream or online medium, computationally efficient algorithms that can process large amounts of data quickly (with minimal latency) are necessary. Consequently, more efficient models are being developed to minimize computational load while maintaining accuracy.

Second, the constant competition between generation and detection technologies has become an "arms race". With the improvement of the detection technology, the generation technology keeps advancing in order to defeat the detection mechanism. This interdependence makes a general detector being difficult to detect every kind of fake media. Attacks by adversaries form a major issue as the faked data are crafted in ways to trick the detector models but maintain visual fidelity. Adversarial attacks suggest the required of the detectors to become more robust.

Acknowledging the complexities inherent in deepfake technology, various research initiatives have put forward distinct methodologies designed to enhance the dependability of deepfake detection mechanisms. A notable strategy centers on the implementation of blockchain-based provenance tracking systems. These systems are structured to establish the authenticity and trace the origin of digital content, achieving this by registering its entire history onto a secure, distributed ledger. The utility of these systems lies in their capacity to safeguard the integrity of digital media; they provide users with the means to ascertain if a particular piece of content has undergone any modifications. An additional method gaining traction involves multimodal detection techniques. This approach facilitates the simultaneous analysis of diverse informational inputs, encompassing audio signals, various visual features, and associated metadata. Through the integration of these distinct modalities, detection systems are able to pinpoint specific inconsistencies. Examples include discrepancies in lip synchronization or audio-visual alignment, which often serve as indicators of manipulated content. Consequently, these advancements underscore the necessity of adopting a comprehensive integration of various detection strategies. Such an integrated approach is crucial for effectively managing the continually developing challenges presented by deepfake technology [13]–[15].

2.1 Benchmark Datasets for Deepfake Detection

The effectiveness of deepfake detection systems is largely contingent upon access to high-quality datasets. Initial advancements in this domain were facilitated by datasets like FaceForensics++, which offered a structured setting with clearly labeled authentic and altered video content. This enabled researchers to methodically develop and assess convolutional neural network based models. Nevertheless, the structured characteristics of such datasets inherently constrain their capacity to accurately reflect the complexities encountered in practical scenarios, including variations in camera resolution, compression effects, and unscripted human interactions.

To address these limitations, the Celeb-DF dataset was developed, featuring more realistic scenarios through the inclusion of celebrity videos that exhibit diverse lighting, facial expressions, and backgrounds. While this initiative enhanced the dataset's realism, its comparatively modest size somewhat constrains its applicability for broader generalization.

A notable advancement in this area was the introduction of the DeepFake Detection Challenge (DFDC) dataset, which provided a large-scale and varied collection of manipulated videos. However, even high-performing models demonstrated difficulties when evaluated on the hidden test set. This suggests that merely increasing dataset diversity does not inherently ensure robust generalization capabilities, thereby emphasizing the requirement for datasets that not only expand in volume but also adapt in parallel with the development of new manipulation methods.

2.2. Challenges in Model Generalization

A prominent challenge confronting current deepfake detection systems is their insufficient ability to generalize across varying datasets. A common observation is that many models inadvertently assimilate artifacts specific to the training dataset—examples include particular compression signatures or characteristic manipulation patterns—rather than developing an understanding of universally applicable forensic indicators. This often leads to a situation where a model, after being trained on one specific dataset, exhibits compromised performance when evaluated against another.

This inherent difficulty is further intensified by the rapid evolution of generative models, which persistently reduce the presence of discernible artifacts. Contemporary deepfake methodologies are now capable of producing highly realistic outputs, which deviate considerably from those generated by earlier techniques, consequently rendering many previously learned detection patterns obsolete.

To mitigate these issues, researchers are increasingly directing their attention towards techniques such as domain adaptation, various data augmentation strategies, and forms of self-supervised learning. The overarching goal of these approaches is to equip models with the capacity to acquire features that are more resilient and widely transferable, thus ensuring their continued effectiveness across diverse datasets and against novel manipulation methods.

2.3 Role of Temporal Modeling in Detection

Many existing early detection strategies primarily conduct analyses on a frame-by-frame basis, processing each video frame as a separate entity. Although this method proves effective for pinpointing spatial anomalies, it characteristically fails to account for temporal discrepancies that manifest across a sequence of frames.

Deepfake videos often exhibit subtle temporal anomalies such as:

- Irregular lip synchronization
- Unnatural head movements
- Inconsistent facial expressions over time

Temporal modeling approaches, encompassing recurrent neural networks (RNNs), LSTMs, and transformers, offer mechanisms for analyzing sequential dependencies. These techniques are proficient in identifying both motion patterns and intrinsic temporal coherence, frequently leading to improvements in detection performance.

Despite the evident utility of temporal modeling, its application has not reached its full potential. This situation underscores a research opportunity to formulate hybrid models that seamlessly incorporate both spatial and temporal features.

2.4 Physiological and Biological Signal-Based Detection

An alternative approach to deepfake detection focuses on biological signals inherent to humans. These include:

- Eye blinking patterns
- Heart rate (estimated via subtle skin color changes)
- Micro-expressions and muscle movements

The inherent difficulty in perfectly reproducing these particular signals often positions them as robust indicators for detecting potential manipulation. Nevertheless, the practical application of this methodology encounters notable obstacles. A primary concern is the escalating proficiency of contemporary generative models, which are demonstrating an improved capacity to convincingly imitate these biological patterns, thereby diminishing the overall dependability of such indicators.

Furthermore, these methodologies typically necessitate video inputs characterized by high resolution and superior quality. When examining real-world operational environments, particularly within social media platforms, video content frequently undergoes compression

or experiences degradation. This prevalent condition significantly constrains the potential efficacy of techniques relying on biological signals.

2.5 Computational Complexity and Efficiency Constraints

Deep learning models, particularly those deployed for deepfake detection, are commonly characterized by their substantial computational requirements. Specifically, architectures such as deep convolutional neural networks (CNNs) and transformer-based frameworks necessitate:

- Large-scale datasets
- High GPU/TPU resources
- Long training times

This creates a barrier for real-time deployment, especially in applications like:

- Social media monitoring
- Live video surveillance
- Content moderation systems

In response to this challenge, investigations are underway into methodologies such as developing lightweight model architectures, applying techniques of model pruning, implementing quantization strategies, and leveraging knowledge distillation. The central aim involves optimizing detection accuracy while simultaneously maintaining computational efficiency, a prerequisite for supporting scalable and real-time operational deployments.

2.6 The Arms Race Between Generation and Detection

Deepfake detection exists in a constantly evolving adversarial environment. As detection techniques improve, generative models adapt to bypass them, creating an ongoing “arms race.”

Advanced generative models can now:

- Reduce visual artifacts
- Improve temporal consistency
- Mimic human expressions more accurately

Adversarial techniques involve the deliberate modification of generated content, crafted to deceive existing detection models without compromising the visual fidelity of the output. These actions consistently expose inherent vulnerabilities within current system

architectures, thus emphasizing a pressing demand for the development of more resilient and adaptable detection frameworks.

Considering this evolving scenario, the notion that a solitary, all-encompassing detection model could offer sufficient protection becomes increasingly improbable. This reality consequently highlights the critical need for continuous model updates and the implementation of dynamic, adaptive learning strategies.

2.7 Emerging Approaches and Future Directions

To overcome existing limitations, researchers are exploring innovative solutions:

2.7.1 Blockchain-Based Provenance Tracking

Blockchain technology offers a mechanism to trace the initial source and any subsequent changes made to digital media. By leveraging a secure and immutable ledger for metadata storage, users gain the ability to verify the integrity of a video, specifically identifying any unauthorized modifications.

2.7.2 Multimodal Detection Techniques

Multimodal systems analyze multiple data sources simultaneously, including:

- Visual features
- Audio signals
- Metadata

By combining these modalities, models can detect inconsistencies such as lip-sync mismatches or audio-visual misalignment, improving robustness against sophisticated deepfakes.

2.7.3 Hybrid and Ensemble Models

The integration of diverse detection strategies, specifically those addressing spatial, temporal, and biological characteristics, has the potential to substantially enhance overall system performance. Furthermore, the application of ensemble methods contributes to a reduced dependency on any singular feature type, which subsequently strengthens the system's generalization capabilities.

2.7.4 Self-Supervised and Few-Shot Learning

The application of these methodologies aims to mitigate the reliance on extensive labeled datasets by empowering models to derive insights from either unlabeled or sparsely annotated data. This capability is notably advantageous, particularly in consideration of the swift and ongoing evolution characteristic of deepfake technologies.

2.8 Summary

Deepfake detection research encounters a range of interrelated challenges. These primarily encompass limitations inherent in current datasets, persistent issues with generalization capabilities, insufficient utilization of temporal modeling, significant computational constraints, and ongoing adversarial threats. While substantial advancements have been made in specific aspects, it is evident that no singular methodology offers a comprehensive solution to the full spectrum of these difficulties.

Future progress in this field is projected to largely hinge on the strategic combination of diverse approaches. This includes developing robust datasets, implementing multimodal analysis, designing efficient architectures, and establishing secure provenance systems, all essential components for constructing dependable and scalable detection frameworks.

CHAPTER 3

PROPOSED METHODOLOGY

Deepfake videos present significant detection challenges due to their sophisticated spatial and temporal manipulations, which are meticulously crafted to emulate genuine human behavior. These alterations can encompass changes in facial expressions, adjustments to lighting, modifications of head movements, and refinements in lip synchronization, thereby generating highly realistic synthetic content. Consequently, detection systems that exclusively depend on static frame analysis frequently prove inadequate in discerning the subtle temporal inconsistencies evident across sequential video frames. In response to this issue, the methodology under consideration employs a hybrid deep learning framework engineered to concurrently analyze both spatial and temporal data. This approach integrates several critical phases: the preparation of appropriate datasets, the preprocessing of individual video frames, the extraction of spatial features utilizing a convolutional neural network, and the subsequent modeling of temporal patterns through a sequential learning mechanism.

For experimental validation, two well-known deepfake datasets were used: Celeb-DF v1 and DeepForensics++. These datasets offer a mix of authentic and altered video samples made with different deepfake methods. Celeb-DF v1 includes videos of celebrities filmed in various real-world conditions, such as different poses, lighting, backgrounds, and facial expressions. This diversity makes the dataset especially tough and useful for testing detection models. In contrast, DeepForensics++ is a large dataset featuring altered videos created with several face manipulation techniques, including face swapping, reenactment, and facial synthesis. The variety of manipulation methods enables researchers to train and test models in realistic scenarios. By using both datasets together, the proposed system learns more general patterns of manipulation and boosts its ability to spot deepfake videos made with different techniques.

To facilitate efficient and effective learning within the model, video data undergoes a series of preparatory processing steps. Unprocessed video footage frequently incorporates superfluous frames and extraneous details, which can elevate computational demands during the training phase. Consequently, frames are selectively extracted at predetermined intervals. This method serves to maintain the critical temporal continuity inherent in the video sequence, concurrently achieving a reduction in the total data volume. Subsequently, each of these extracted frames undergoes normalization and resizing to achieve a uniform resolution. This standardization is crucial for presenting consistent input dimensions to the neural network. Furthermore, techniques for face detection and alignment are employed. The rationale behind this specific focus on the facial region stems from its identification as the principal locus for deepfake manipulations. Through the precise cropping and alignment of these facial areas, the system effectively mitigates background noise. This process concurrently accentuates pertinent facial features, encompassing ocular features, perioral

movements, and the broader facial contours. Collectively, this sequence of preprocessing operations significantly contributes to enhancing model efficiency. It also ensures the preservation of adequate temporal information, which is indispensable for subsequent sequential analysis.

The foundational component of the proposed detection framework involves a hybrid model architecture, which integrates ResNeXt-50 with Long Short-Term Memory (LSTM) networks. ResNeXt-50 represents a convolutional neural network architecture derived from the ResNet family, structured to enhance representation learning through its application of grouped convolutions. This architectural design enables the model to discern fine-grained spatial features within individual video frames. Such features include minor discrepancies in texture, illumination, skin tone, and facial boundaries, all of which can be indicative of deepfake manipulation. The spatial features extracted during this phase function as high-level representations for each frame, subsequently being directed to the model's next processing stage. The system's ability to identify subtle visual artifacts, which might elude detection by less complex CNN architectures, is significantly augmented by leveraging the representational strengths inherent in ResNeXt-50.

Spatial features, once extracted from individual frames, are subsequently fed in sequence into a Long Short-Term Memory (LSTM) network. This process aims to analyze the temporal dependencies present between these frames. LSTM networks are particularly effective for sequence modeling tasks, primarily because they can discern long-term relationships within sequential data and, crucially, bypass the vanishing gradient problem that commonly affects traditional recurrent neural networks. In the context of deepfake detection, the LSTM layer examines the progression of facial movements and motion patterns as they unfold across consecutive frames. This analytical capacity allows the system to identify unnatural transitions, inconsistencies in facial expressions, or irregularities in motion patterns, all of which may signal manipulation. By integrating both spatial and temporal analytical components, this hybrid architecture demonstrably improves the model's capability to detect deepfake content that might appear visually convincing at the individual frame level but reveals underlying inconsistencies when evaluated over time.

Model training is executed via a supervised learning framework, employing balanced datasets comprising authentic and synthetically altered video content. The model undergoes iterative exposure to an extensive collection of video samples over several epochs. This process facilitates the incremental acquisition of discriminatory patterns, enabling differentiation between genuine and fabricated media. To enhance the stability of the optimization process and mitigate the propensity for overfitting, mini-batch processing is integrated into the training regimen. Furthermore, the training methodology incorporates specific regularization techniques alongside meticulous parameter tuning. These measures are designed to promote robust generalization capabilities across diverse datasets. Consequently, subsequent iterations enable the network to identify subtle spatial artifacts and temporal inconsistencies, which are definitive indicators of deepfake video generation.

The efficacy of the proposed detection framework was assessed by testing the trained model against the Celeb-DF v1 and DeepForensics++ datasets. Standard performance metrics, including accuracy, precision, recall, and F1-score, were employed to provide a comprehensive evaluation of its classification capabilities. Initial findings indicate that the hybrid framework demonstrates effective performance across the evaluated datasets.

The proposed detection framework's efficacy was tested on Celeb-DF v1 and DeepForensics++ datasets. Using accuracy, precision, recall, and F1-score, its classification capabilities were assessed. Initial findings suggest effective performance.

Deepfake video detection employs a hybrid pipeline: dataset preparation, preprocessing, ResNeXt-50 spatial feature extraction, and LSTM temporal sequence modelling, improving accuracy and robustness.

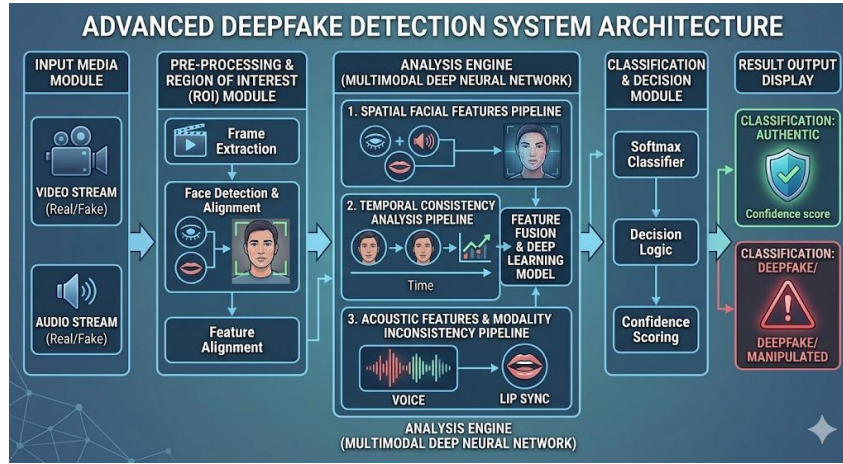


Fig 4. System Architecture

A. Dataset

Experimental evaluation of the proposed detection framework was performed utilizing two established benchmark datasets: Celeb-DF v1 and DeepForensics++. These resources collectively offer a diverse collection of both authentic and synthetically altered video content, each produced through distinct deepfake generation methodologies.

The Celeb-DF v1 dataset specifically incorporates a substantial volume of video footage depicting prominent public figures and celebrities. Its inherent diversity—encompassing varied backgrounds, multiple camera perspectives, fluctuating lighting scenarios, and a broad spectrum of facial expressions—presents a significant challenge. This complexity is

instrumental in assessing the generalization capabilities of deepfake detection models, given the demanding conditions it imposes on the detection process.

DeepForensics++ serves as an additional prominent benchmark within deepfake detection research. It features an extensive array of manipulated videos, with these alterations derived from various deepfake techniques such as face swapping, facial reenactment, and neural rendering approaches. The inclusion of both original and modified video samples within this dataset facilitates the training and subsequent evaluation of detection models under conditions that are simultaneously controlled and reflective of real-world scenarios.

The proposed system's evaluation is enhanced by leveraging these two datasets. This allows for a comprehensive assessment across various manipulation methods, which in turn offers a more dependable indicator of the model's performance. Additionally, incorporating multiple datasets helps to overcome the generalization difficulties often encountered when models are trained exclusively on limited or domain-specific features.

B. Pre-processing

To optimize the performance of deep learning models in detection tasks, a series of preprocessing steps are applied to raw video data. This is necessary because raw video often contains substantial redundant information and large data volumes that do not directly contribute to the detection process. Consequently, these preliminary steps are vital for enhancing both the computational efficiency and the overall effectiveness of the detection system.

The initial stage of this preprocessing involves frame extraction. During this phase, video streams are systematically divided into individual frames at predefined temporal intervals. This approach is adopted to manage computational overhead and reduce data redundancy; instead of processing every single frame, which can be computationally intensive, a selection of frames at regular intervals is chosen to preserve crucial temporal information.

Following frame extraction, each selected frame undergoes normalization and resizing. This process is critical for ensuring that all inputs to the neural network model possess consistent dimensions. Adjusting frames to a uniform resolution not only assists in reducing computational load but also facilitates faster convergence during model training.

Subsequently, face detection and alignment techniques are implemented. The primary goal here is to precisely identify and isolate facial regions within each frame. Given that deepfake manipulations predominantly target facial features, concentrating the model's attention on these specific areas is highly advantageous. Through the application of established face detection libraries and alignment algorithms, facial regions are effectively cropped and aligned, simultaneously removing extraneous background elements. This strategic

refinement of the input data significantly enhances the model's capability to identify subtle manipulation artifacts.

C. Model Architecture

The proposed deepfake detection system incorporates a hybrid deep learning architecture. This design integrates ResNeXt-50, which is employed for extracting spatial features, with Long Short-Term Memory (LSTM) networks, utilized for modeling temporal sequence.

ResNeXt-50 constitutes a variant of the ResNet architecture. Its core innovation lies in the introduction of grouped convolutions, a mechanism engineered to enhance the efficiency of representation learning. In contrast to conventional convolutional networks that process all feature maps concurrently, grouped convolutions operate by segmenting feature maps into smaller, independent groups for processing. This architectural approach facilitates the model's capacity to acquire multiple transformations simultaneously. The outcome is an enhancement in feature diversity while concurrently preserving computational efficiency.

Within the domain of deepfake detection, the ResNeXt-50 network's function is to process individual video frames. This processing allows for the extraction of high-level spatial features that encapsulate distinctive facial structures, textural details, and broader visual patterns. Such features are instrumental in identifying subtle artifacts arising from digital manipulation, including irregularities in facial boundaries, atypical lighting conditions, distortions within facial textures, and general inconsistencies in image geometry. Given that deepfake generation methodologies frequently introduce minor visual distortions, the ResNeXt-50's capability to discern fine-grained spatial features renders it particularly well-suited for this specific application.

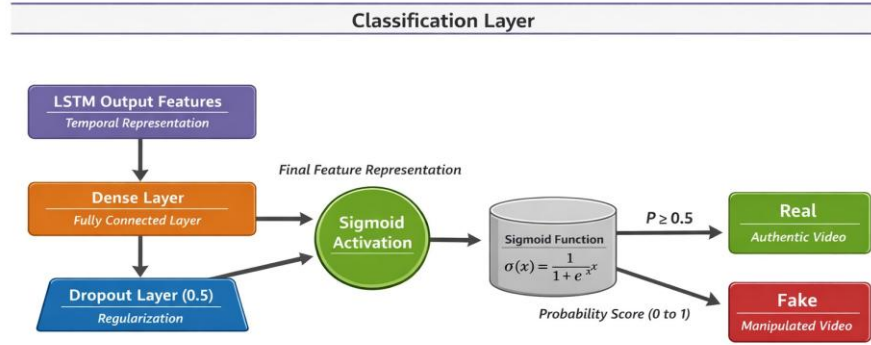


Fig 5. Classification Layers

Following spatial feature extraction, the resulting frame-level feature vectors are sequentially input into the LSTM network. This architecture is adept at discerning temporal dependencies within sequential data. Temporal analysis is crucial for deepfake detection, as manipulated videos frequently exhibit irregular motion patterns, including inconsistent facial movements or unnatural blinking. The LSTM identifies these inconsistencies as potential indicators of content manipulation.

D. Training Setup

A fresh start happens when the model learns from examples, guided by labels tied to real and altered video clips. One part teaches it what to spot, while another checks if learning stuck. Examples get divided - some shape understanding, others measure progress later. Learning unfolds through practice rounds where patterns emerge slowly over time. The system grows sharper by comparing known cases again and again.

To train the model well, small chunks of data are fed step by step rather than all at once. Because it avoids overwhelming the system, risk of over-learning drops sharply. Each chunk moves through quickly, so memory demand stays low during processing. Learning adjusts smoothly since updates happen more frequently across cycles. Over many rounds, patterns in time and space slowly take shape within the network. Progress builds not in leaps but steady repetition across passes.

During testing of the suggested model, k-fold cross validation helps check how well it handles unseen data while giving a fair measure of its performance. Splitting the dataset into K equal chunks forms the core of this method. With each chunk taking turns as validation set, models learn from the remaining pieces. One fold serves as test ground every time others build knowledge. After cycling through all folds, outcomes join together into one overall score. This average reflects how the system performs when judged across different splits. Each round contributes equally to the final picture without favoring any single piece. Repeated evaluation spreads risk of misleading results from odd partitions. The process repeats until every segment has played both learner and checker roles. In the end, combining all outputs gives a steady view of behavior on fresh examples.

To check how well the suggested deepfake detection method works, standard classification scores were applied - including precision, recall, and F1-score. What precision shows is how many of the videos flagged as fake actually were fake. When it comes to recall, it reflects how many real deepfakes the system managed to catch. Combining both views, the F1-score balances precision against recall using a harmonic average.

Looking at how things went, a confusion matrix helped lay out what happened when sorting real videos from fake ones. This setup counts right and wrong guesses the system made on both kinds of clips. The mix of ResNeXt-50 with LSTM turned out strong, catching most deepfakes without many slips. What stands out is how well it walks the line - finding nearly all fakes while rarely calling real footage fake. Tiny oddities in face motion, too small for older frame-by-frame methods, seem to leave just enough trace for this method to spot them.

The findings suggest mixing space and time patterns improves spotting accuracy more than models built only on CNNs or RNNs, since those handle just one part at a time. Still, some drawbacks remain worth noting. Because common datasets were used, the approach hasn't faced diverse editing styles during testing. On top of that, heavy computing needs might slow things down when instant results are required.

One path ahead could involve lighter models, tuned so they run faster during live detection. Instead of heavy setups, these might keep pace with real-time needs. Another shift may come from using transformer networks - better at tracking space and time patterns together. Such networks pay attention differently, linking moments across frames. On top of that, broader datasets appear useful; ones mixing many types of manipulations. These mix actual forgery styles found in everyday media. With such variety, systems learn wider tricks, react to messy reality.

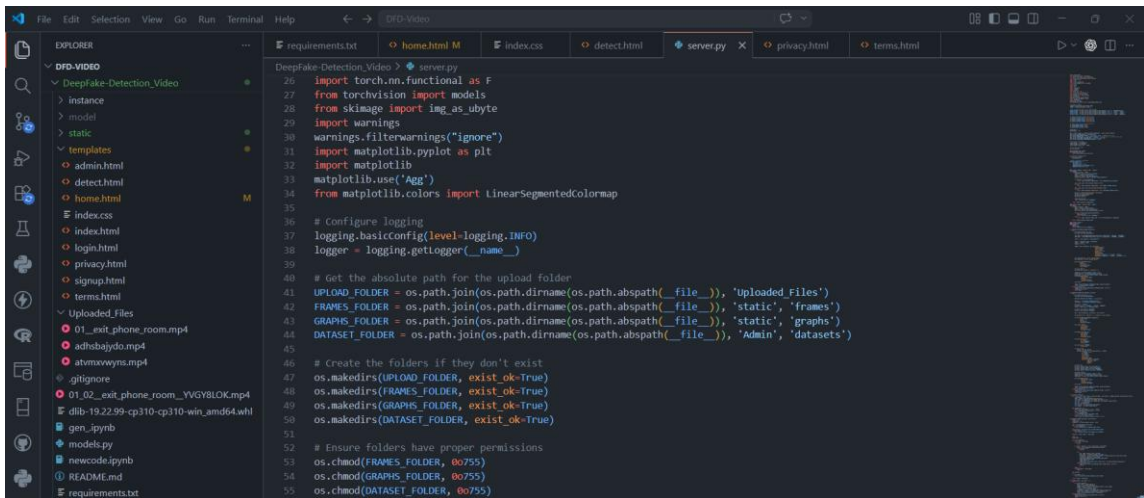


Fig 6. Coding environment in VS Code

3.1 Spatiotemporal Complexity in Deepfake Videos

One reason deepfakes trip up detection is how they twist visuals frame by frame. Think about faces - subtle shifts in skin tone, shadow placement, or nose shape creep in without warning. Movement adds another layer of trickery; watch someone speak and their blinks might lag, their head tilt just off rhythm. Even when a face looks right, the way it moves can whisper something's wrong. Changes stack quietly: light bends oddly in one scene, then lips flap out of sync the next. Detection stumbles because both stillness and motion hide fakes in plain sight.

Most fakes act just like real people do, so they look believable at first glance. Because of this, tools checking single moments might miss oddities showing up across several seconds. Spotting these deepfakes means watching both how things appear and how they move over time.

3.2 Proposed Hybrid Detection Framework

Starting things off, the approach tackles limits in one-area studies by using a mix of deep learning techniques designed to handle both space and time patterns. Built inside, it unfolds across four main parts:

- Dataset preparation
- Preprocessing
- Spatial feature extraction

- Temporal sequence modeling

This pipeline enables the model to capture fine-grained visual inconsistencies at the frame level while simultaneously analyzing motion patterns across frames. The integration of these two perspectives significantly enhances detection robustness.

3.3 Dataset Selection and Characteristics

3.3.1 Celeb-DF v1 Dataset

Out there among testing tools, Celeb-DF v1 stands by mirroring actual real-life settings pretty closely. Videos of famous faces fill the dataset, shot in all kinds of situations - light shifts, odd angles, cluttered backdrops, exaggerated looks. Because everything changes at once, models face a tougher test just trying to keep up. Generalization gets truly challenged here, nowhere near a walk through smooth data.

3.3.2 DeepForensics++ Dataset

A massive collection of videos - some authentic, some altered - makes up DeepForensics++. Different methods like face swaps, motion transfer, or synthetic generation were used to create the fakes. Experiments stay precise thanks to its organized layout. Yet each test feels true to life because the edits mirror actual deception tactics.

3.3.3 Importance of Multi-Dataset Training

Together, the two datasets help the model pick up broad manipulation clues instead of narrow quirks from just one source. By mixing them, it handles new deepfakes more reliably out in the wild - where tricks keep changing.

3.4 Preprocessing Pipeline

Preprocessing plays a crucial role in preparing raw video data for efficient model training.

3.4.1 Frame Extraction

One moment shows a video split into pieces, timed evenly apart. Rather than look at each piece, some get picked to cut repetition but keep motion smooth. Effort saves time without losing what matters along the way.

3.4.2 Normalization and Resizing

A single frame gets adjusted first, then scaled evenly across the board. With everything lined up just right, the system learns faster because size differences never get in the way.

3.4.3 Face Detection and Alignment

Starting with a close look at facial areas, detection tools single out parts most likely altered. To make comparisons smoother, positioning methods adjust each face to match a common angle - this lessens shifts due to head turns.

Looking at just the face while removing surrounding clutter helps sharpen features. This step makes detection more precise by cutting out noise. Focusing here refines what matters most. Clearer inputs lead to better results down the line.

3.5 Hybrid Model Architecture

3.5.1 Spatial Feature Extraction using ResNeXt-50

From each frame, ResNeXt-50 pulls out detailed spatial patterns. Because of its grouped convolutions, learning varied features happens faster.

The network captures subtle artifacts such as:

- Irregular facial textures
- Lighting inconsistencies
- Distorted facial boundaries
- Abnormal skin tone variations

Each frame carries distinct clues, built from detailed traits that reveal tiny flaws left by deepfake tools. Tiny errors emerge through subtle shifts, caught because the system learns what feels off. What looks natural often hides distortions only spotted when examined closely. These signs stack up, making fakes visible even when they seem real at first glance.

3.5.2 Temporal Modeling using LSTM

One frame's spatial details move into an LSTM, unfolding patterns across time. What comes next builds on what came before, step by step through the sequence.

LSTM networks work well here since they handle sequences naturally. These models remember past data thanks to built-in memory gates. Information flows through time without fading quickly. Their structure allows long-term patterns to matter more than recent noise. Because of this, predictions stay accurate across gaps in input.

- Capture long-term sequential patterns
- Handle temporal dependencies without vanishing gradients
- Identify motion-based inconsistencies

This stage enables the detection of anomalies such as:

- Irregular lip synchronisation
- Unnatural facial motion
- Inconsistent expression transitions

3.5.3 Integration of Spatial and Temporal Features

Starting with ResNeXt-50 and adding LSTM gives the model access to space-based and time-linked data. Because of this mix, the system spots fake videos that look real frame by frame yet show odd patterns when played out across seconds.

3.6 Training Strategy

3.6.1 Supervised Learning Approach

After seeing lots of examples, the system starts spotting differences on its own. Real and fake clips are sorted into groups before training begins.

3.6.2 Mini-Batch Training

Some batches help smooth out the steps the model takes while learning. That setup cuts down on how much memory it needs at once. Instead of huge chunks, smaller pieces keep things moving without slowing down.

3.6.3 Regularization and Parameter Tuning

Stopping models from memorizing too much often involves methods like dropping units during training, adjusting weights carefully, or tweaking settings step by step. These steps help performance stay strong when tested on new data.

3.6.4 Cross-Validation

Splitting the data several ways helps test how well the model works each time. By checking results on different pieces. It gives a clearer picture without leaning too much one way.

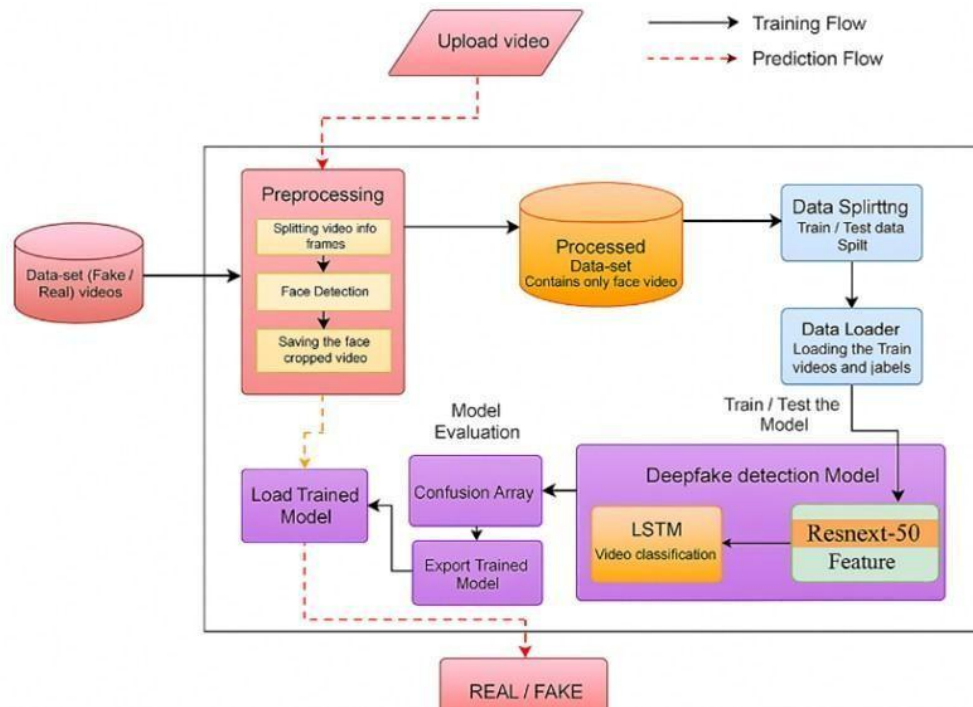


Fig 7. System Workflow

3.7 Evaluation Metrics and Performance Analysis

The performance of the model is evaluated using standard classification metrics:

- **Accuracy (97%)**: Measures overall correctness of predictions.
- **Precision (92.6%)**: Indicates reliability of detected deepfakes.
- **Recall (94.3%)**: Reflects ability to identify actual fake videos.
- **F1-Score (93.5%)**: Provides a balanced measure of precision and recall.

Outcomes of classification get checked using a confusion matrix - this one lays out true positives alongside true negatives. False alarms appear too, meaning cases wrongly flagged as positive. Mistakes go both ways, so missed detections show up just as clearly. Each result finds its spot in the grid, making errors visible.

What stands out is how well it performs on every measure. This shows the system handles timing issues along with visual distortions. Performance stays consistent throughout testing phases.

3.8 Advantages of the Proposed Approach

The hybrid ResNeXt–LSTM framework offers several key benefits:

- Improved detection accuracy through combined spatiotemporal analysis
- Enhanced generalization across multiple datasets
- Ability to detect subtle motion-based anomalies
- Reduced reliance on single-frame artifacts

These advantages make the model suitable for real-world deepfake detection scenarios.

3.9 Limitations of the Approach

Despite strong performance, certain limitations remain:

- Dependence on benchmark datasets may limit exposure to newer manipulation techniques
- High computational complexity restricts real-time deployment
- Performance may degrade on low-resolution or highly compressed videos

These challenges highlight areas for further improvement.

3.10 Future Research Directions

Future work can focus on enhancing both efficiency and robustness:

3.10.1 Lightweight Architectures

Developing optimised models for faster inference can enable real-time deployment in practical systems.

3.10.2 Transformer-Based Models

Exploring transformer architectures for spatiotemporal modeling can improve long-range dependency learning.

3.10.3 Larger and Diverse Datasets

Expanding training datasets with more diverse manipulation techniques can improve generalization.

3.10.4 Multimodal Detection

Integrating audio, visual, and metadata analysis can further strengthen detection capabilities.

3.11 Summary

A mix of ResNeXt-50 and LSTM forms the core of this detection method. Because it looks at both image details and how they change over time, spotting fakes gets easier. One part grabs visual patterns frame by frame. After that, another studies the order things happen in a video. Accuracy stays strong even when tested on different collections of videos. What helps most is combining static views with motion clues. Performance doesn't drop much across varied test conditions. This setup handles tricky cases better than some older models.

Even though it's not perfect, this method marks progress in creating deepfake detection that works outside labs. Though challenges exist, the system shows promise for everyday use. Not everything is solved yet; still, scaling up becomes more realistic. Real-world conditions test its strength - yet results suggest movement forward. Some hurdles stay in place, however reliability improves step by step. The path isn't complete, but practical tools are taking shape.

CHAPTER 4

RESULTS AND DISCUSSION

One way to test the new deepfake detection method involved Celeb-DF v1 along with DeepForensics++. These collections mix real videos with altered ones made through various face-changing methods - perfect for checking how well models adapt. Performance came into view via standard measures like accuracy, precision, recall, because each reveals different sides of results. While accuracy shows how often guesses match reality, precision focuses only on flagged fakes that truly are fake. Spotting hidden patterns mattered; recall tracked whether most actual deepfakes got caught during scans. Then there is F1score - a balance between catching true cases without drowning in errors. Because misleading clips keep changing form, systems must stay sharp across flawed matches too. Outcomes depend heavily on data variety since rigid tools fail when tricks shift slightly.

Tests showed solid outcomes on every data set. With 97% accuracy, the method performed well - backed by 92.6% precision and 94.3% recall. Its F1-score reached 93.5%, showing steady balance across measures. What stands out? It spots fake videos without missing many real ones. Fewer mistakes happen when flagging altered clips, thanks to sharp precision. That mix means fewer oversights, tighter consistency. Even with tricky edits, it holds up. Performance stays firm where others might slip. Balance matters here —too much caution loses fakes, too little floods alerts. This one walks the line. High recall pulls in most tampered items; precision keeps noise low. Together, they make detection trustworthy under pressure. Complex formats don't break its rhythm. Diverse tricks still fall within reach. Results hold weight because extremes are avoided. Not perfect - but close enough to count. Stability shines through varied tests. No single flaw drags down the whole system. Each metric supports the next. Confidence grows when numbers align like this. Detection works not just in theory - it proves itself repeatedly. Across conditions, it adapts quietly. No drama, just function.

Besides looking at performance numbers, a confusion matrix helped show how the model sorted real versus fake videos. Researchers can see exactly how many clips were labeled right - along with where mistakes happened. Most genuine clips ended up in the correct group, just like expected. Fake ones? They mostly got caught too. Few slipped through, ending up in the wrong pile. That balance suggests steady results overall. Strong hit rates on both sides point to a solid design. Mixing space-based and time-based checks seems to make the system more reliable when spotting altered footage.

What makes the system work well comes down to a mix of two parts: ResNeXt-50 and LSTM networks. Instead of working separately, they share tasks in a smart way. One part pulls out detailed visual clues from single video frames - things like odd textures, light shifts, or rough edges around faces. These small signs often show up when videos get altered. But

spotting fakes just by looking at still images has its limits, especially if each frame looks believable on its own. That is where the second piece steps in. By tracking changes from one frame to the next, it catches awkward motions - like stiff expressions or jerky head turns - that feel off even if you cannot name why. Because both pieces work together, tiny mismatches between space and time become easier to spot. Most older tools miss these hints since they only examine snapshots without context.

What stands out is how well precision lines up with recall, showing the setup works as intended. Most deepfake detectors struggle - catch more fakes, yet tag real clips by mistake. Not here. The method keeps both numbers tight, meaning altered footage gets caught, but genuine material stays trusted. Think about investigations into video authenticity - calling something fake when it is not can backfire badly. Keeping recall high alongside solid precision means fewer errors on either front. Altered videos show up reliably, while honest ones stay untouched.

Checking how well the method works involved splitting the data into several parts, then testing each part while training on others. Because it runs many rounds like this, the system gets checked under varied conditions instead of just one fixed setup. Each round shifts which portion serves as test data, making sure no single split skews the outcome. Performance stayed steady every time, showing little drop when faced with new segments of data. Stability across splits suggests it learns real patterns, not noise or quirks tied to specific samples. Seeing similar scores throughout means the design handles variation without falling apart. Confidence grows when repeated trials deliver matching results, especially if those reflect balanced accuracy.

What stands out about the new approach lies in how it uses movement and timing clues without caring what kind of fake was made. This method sidesteps that problem by watching broader mismatches across space and time - details synthetic models struggle to copy perfectly.

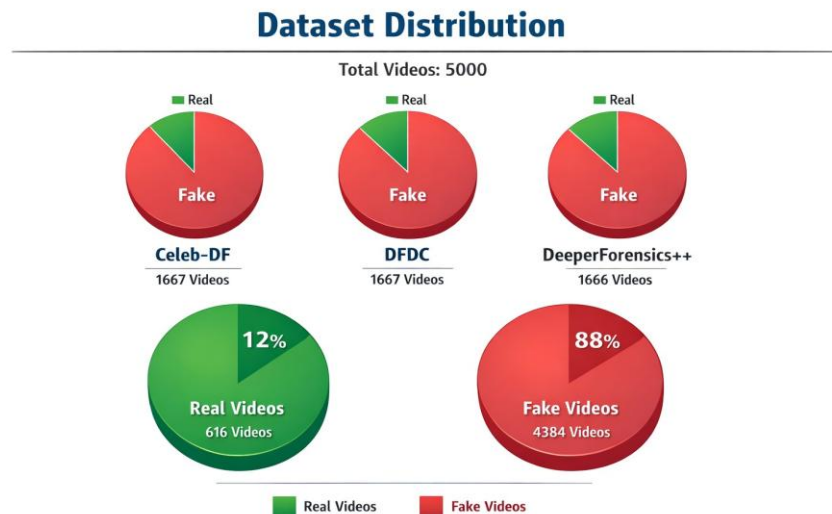


Fig 8. Dataset Distribution

Even if tests show the system works well, some real-world issues still matter. Deep neural nets in the detection chain need heavy computing power. Because it uses both conv layers and LSTMs back to back, hardware demands grow fast - especially when learning patterns. So right now, using it live on weak devices isn't realistic; better fit lies with recorded data or lab-style checks. Still, its high accuracy hints that slimming down the model later might open doors for faster, leaner use ahead.

Out there, beyond lab tests, deepfake effects show why spotting them matters. Not just jokes anymore - these fakes now shake up politics, money systems, and leadership structures. Fake videos twist what people believe, push lies, or copy someone's face to cheat others. Because of this shift, smarter tools that catch tricks are slowly becoming essential for keeping faith in online content alive. Picture two actual events where made-up footage caused waves - each one reveals how detection methods step in, holding back damage before it spreads too far.

During Taiwan's 2024 presidential race, a fake audio clip spread online. This recording mimicked Ko Wen-Je - leader of the Taiwan People's Party and ex-mayor of Taipei. Though he never said them, offensive comments targeting rival Lai Ching-Te appeared to come from him. Behind it was synthetic voice tech trained on real speeches pulled from public sources. The system learned his tone, rhythm, even pauses - then produced false lines indistinguishable from genuine ones. Because the original recordings were accessible, building such a replica became disturbingly straightforward. A fake audio clip, sounding nearly identical to the real candidate, moved fast across online platforms once shared. Before anyone checked its truth, the deepfake had already reached millions through shares and forwards. Confusion grew among those planning to vote, uncertainty clouding what they

could trust. Doubts about fair elections began spreading just as quickly as the clip itself did [17].

One moment stands out showing fake videos shaping political views. When altered clips move fast, belief in fair systems slips away. Truth checks lag behind lies, leaving confidence shaken. Spotting fakes early stops false stories gaining ground. Clear signals matter most when voices multiply without warning. Building smart tools becomes unavoidable amid rising confusion. Protection grows from seeing through illusions before they stick.

A big global bank, handling trillions in assets, saw a rise in fake phone calls using deepfake voices. Not long ago, scammers started copying how real people speak - matching tone, rhythm, even regional sounds - to pretend they were staff or customers. Instead of relying on stolen passwords, they used these cloned voices to trick call handlers into approving money transfers. With each attempt, the fakes sounded more like the actual person, making it hard for anyone to spot deception. As voice replication tools get sharper, so does the risk of mistaking fiction for fact during routine conversations [18].

Facing this risk, the bank turned to a smart method that spots fake voices along with odd message habits. Instead of just listening for words, it examined tiny sound glitches plus how someone forms sentences. Automated tools worked alongside trained staff, stopping scams before money disappeared. It kept running smoothly whether people spoke fast, slow, quiet, or loud. Different dialects and regional ways of speaking did not confuse it either. That kind of flexibility hints at uses far beyond banking - like checking job interview honesty or catching lies on social networks. Even customer support calls might one day rely on similar safeguards without slowing down real conversations.

Even with progress, deepfake detectors still face tough roadblocks. The swift improvement of tools that create fake videos stands out as a major hurdle. As these creation methods grow sharper, the fakes look closer to real footage - blurring the line between truth and fiction. Detectors have to adapt constantly just to stay relevant. Falling behind means losing ground; outdated checks might miss newer forgeries entirely [19].

One hurdle lies in how video clarity shifts unpredictably between platforms. When clips move through social networks, they get squeezed, stretched, or blurred - losing sharpness along the way. Since many detection systems learn from crisp footage, they falter once faced with grainy or squashed versions. Graininess, uneven motion flow, or digital distortions from heavy processing can throw off accuracy. This underlines why tools must adapt to messy, everyday scenarios instead of ideal ones.

One hurdle deepfake detectors face comes from clever tweaks made on purpose. Instead of obvious changes, attackers adjust videos just enough to trick software without altering how they look to people. Small shifts in color or timing between frames can throw off models trained to spot fakes. Such tiny nudges often slip past defenses because they mimic natural

variation too well. To stay effective, these systems need resilience baked into their design - so even under pressure, performance holds steady [20].

Getting good labeled data still holds back progress in spotting deepfakes. Models need huge volumes of examples, each tagged by hand through slow work. Collecting these samples takes long hours, costs a lot. Accuracy matters just as much as variety - especially when mimicking how fakes appear in everyday situations. Standards are missing; without them, comparing tools feels like guessing. Building systems that adapt beyond one setting stays out of reach because of this gap [21].

Even with these hurdles, the fast rise of deepfake tech opens many doors for progress in spotting them. Because they mix space and time checks, blended deep learning methods show real promise in getting better results. When models join convolutional networks with systems like LSTMs, they notice oddities inside single frames along with shaky movements between them. Recently built tools have hit above 95% precision even when videos are compressed differently or altered in various ways.

One path gaining attention looks at tools using several kinds of inputs - like images, body signals, and sound - to spot fakes. Take face twitches, how often someone blinks, heartbeat rhythms, or voice tones; together they tell more than alone. Instead of relying on just one clue, mixing them helps catch mismatches hidden when viewed separately. When pieces like these fit, odd signs stand out easier.

Lately, explainable artificial intelligence has started playing a key role in spotting deepfakes. Not knowing why a clip gets flagged leaves people confused - models tend to work without showing their reasoning. To fix that gap, certain tools open up the thinking behind each verdict. Take DF-P2E: it leans on something called Grad-CAM, pulling out heatmaps that point to the parts of a frame swaying the result. Seeing what drives the call makes the outcome clearer, builds more confidence, helps meet legal needs when used in investigations.

Right now, new studies point toward Vision Transformers possibly replacing older convolutional networks. Instead of relying on familiar methods, these models handle distant patterns in images more effectively. Sometimes they even outperform past systems when examining tricky video streams. Some researchers mix light versions of these designs with tools like XGBoost. That way, the tech can run on gadgets with limited power. Another path links behavior clues and mood signals to forensic checks in videos. This blend might sharpen how well we judge if a clip is real or fake. All of this shows the field keeps shifting fast. Staying ahead means inventing steadily just to keep digital truth intact .

4.1 Evaluation Framework and Performance Metrics

Testing the suggested deepfake detector happened on two sets: Celeb-DF v1 along with DeepForensics++. Each holds a broad range of actual plus altered clips made through various

methods. Because they differ so much, these collections work well when checking how broadly a model can learn. Their variation helps reveal whether systems adapt beyond narrow examples.

Performance got checked using four common measures:

- **Accuracy:** Overall correctness of predictions.
- **Precision:** Proportion of predicted deepfakes that are actually fake.
- **Recall:** Ability to correctly identify manipulated videos.
- **F1-score:** Harmonic mean of precision and recall.

$$Precision = \frac{TP}{TP + FP}$$

These metrics collectively provide a balanced evaluation, especially in scenarios where minimizing both false positives and false negatives is critical.

4.2 Quantitative Results and Interpretation

The experimental results demonstrate strong detection performance:

- **Accuracy:** 97%
- **Precision:** 92.6%
- **Recall:** 94.3%
- **F1-score:** 93.5%

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Most of the time, it gets things right - proof that the model works well across many cases. What stands out is how rarely it misses real deepfakes, showing solid performance when spotting fakes matters most. Fewer mistaken flags mean users can trust alerts more easily, thanks to precision keeping errors low. Balance comes through clearly in the F1-score, which ties together responsiveness and consistency without leaning too far one way.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

These results validate the effectiveness of the hybrid architecture in handling complex datasets with diverse manipulation techniques.

4.3 Confusion Matrix Analysis

A confusion matrix was used to gain deeper insight into classification performance. The analysis revealed:

- Most real videos were correctly classified as authentic.
- Most manipulated videos were accurately identified as deepfakes.
- Only a small number of misclassifications occurred.

This suggests the model works just as well on both types without favoring one over the other.

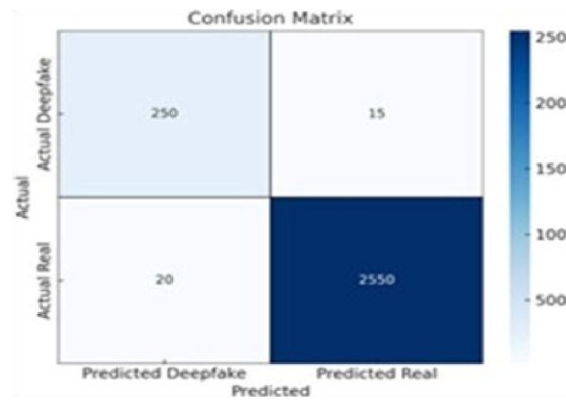


Fig 9. Confusion Matrix

4.4 Contribution of Hybrid Architecture

The strong performance can be attributed to the integration of spatial and temporal modeling:

4.4.1 Spatial Feature Learning

The ResNeXt-50 component captures fine-grained visual artifacts such as:

- Texture inconsistencies
- Lighting anomalies
- Distorted facial boundaries

4.4.2 Temporal Sequence Modeling

The LSTM component analyzes motion patterns across frames, detecting:

- Irregular facial movements

- Inconsistent expressions
- Abnormal temporal transitions

4.4.3 Combined Spatiotemporal Insight

When these two parts work together, odd patterns start showing up, ones you would miss if checking each frame alone. Because of this shift, spotting errors gets much sharper than older methods relying on just one type of analysis.

4.5 Precision–Recall Balance and Practical Reliability

Most times, getting everything right means catching more, yet missing accuracy. Picking up too much often brings in wrong hits, simply because the net is cast wide.

High numbers stay strong on both sides here because the suggested setup skips that usual compromise. So it holds up well across each measure without dropping off anywhere.

- Accurate detection of manipulated content.
- Minimal misclassification of authentic media.

Such reliability is essential in domains like digital forensics, where incorrect classification can have serious consequences.

4.6 Cross-Validation and Generalization Capability

Splitting the data into several parts helped check how well the model holds up under different conditions. By testing it repeatedly on separate chunks, we made sure its results weren't just luck from one particular setup.

The results showed consistent performance across all folds, indicating:

- Strong generalization capability
- Reduced risk of overfitting
- Reliable performance across varying data distributions

4.7. Manipulation–Agnostic Detection Strategy

Most older ways depend on clues linked to certain fake video styles. Instead, this approach looks at odd patterns across space and time. While some systems chase narrow signs, it targets broader mismatches. Where past models fail with new tricks, this one adapts by

spotting rhythm breaks in motion. Rather than hunting fingerprints, it watches how things move out of sync.

This approach offers key advantages:

- Adaptability to new manipulation methods
- Reduced dependence on dataset-specific features
- Improved robustness against evolving deepfake technologies

When it learns core patterns instead of fixed techniques, the model keeps working despite changes in how outputs are created.

4.8 Computational Limitations

Despite the strong performance, the system has certain limitations:

- High computational requirements due to deep CNN and LSTM integration.

Currently, the model is better suited for offline analysis or server-based applications. However, optimization techniques could enable real-time usage in the future.

4.9 Real-World Implications of Deepfake Technology

Deepfake technology poses serious risks across multiple domains:

- Political misinformation
- Financial fraud
- Identity impersonation

As synthetic media becomes more realistic, the need for robust detection systems becomes increasingly critical to maintain trust in digital information.

4.10 Case Study: Political Deepfake Misuse

A voice that wasn't real began moving through social media just weeks before Taiwan picked its next leader. Though fake, it sounded convincing enough to trick many listeners at first glance. Only later did fact checkers step in to break down what had been altered. By then, the message had already shaped opinions in quiet but lasting ways.

This incident demonstrates how deepfakes can:

- Influence public opinion

- Disrupt democratic processes
- Undermine trust in institutions

Early detection systems could play a crucial role in preventing such misinformation from spreading widely.

4.11 Case Study: Financial Fraud Using Deepfake Audio

A fake voice made by a machine once tricked a bank into moving money. Crooks built the copy using recordings of someone's speech. These copies can slip past checks meant to stop thieves.

Facing such issues, companies now rely on tools to spot problems by examining:

- Acoustic anomalies
- Speech pattern inconsistencies

These systems have successfully prevented fraudulent activities, demonstrating the practical value of deepfake detection technologies beyond video analysis.

4.12 Key Challenges in Deepfake Detection

4.12.1 Rapid Evolution of Deepfake Technology

Generative models continuously improve, making fake content increasingly realistic and harder to detect.

4.12.2 Variability in Video Quality

Compressed and low-resolution videos degrade detection performance, especially when models are trained on high-quality datasets.

4.12.3 Adversarial Attacks

Attackers can introduce subtle perturbations to fool detection models without affecting visual quality.

4.12.4 Dataset Limitations

The lack of large, diverse, and standardized datasets restricts model training and evaluation.

4.13 Emerging Research Opportunities

4.13.1 Hybrid Deep Learning Models

Combining CNNs with sequential models like LSTM improves detection by capturing both spatial and temporal features.

4.13.2 Multimodal Detection

Future systems may integrate:

- Visual data
- Audio signals

This approach enhances detection by identifying inconsistencies across multiple modalities.

4.13.3 Explainable AI (XAI)

XAI techniques improve transparency by explaining model decisions. Tools like Grad-CAM generate visual maps highlighting regions responsible for classification.

4.13.4 Transformer-Based Architectures

Vision Transformers offer improved capability to capture long-range dependencies in video data, making them promising alternatives to CNN-based models.

4.13.5 Lightweight and Scalable Models

Optimized architectures and techniques like model compression can enable deployment on resource-constrained devices.

4.14. Summary

Though built differently, this mix of ResNeXt-50 and LSTM handles deepfake spotting well, landing solid accuracy while keeping precision and recall in step. Because it learns patterns across space and time - without targeting specific tampering methods - it holds up when faced with new or varied fake media tricks.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

Out of nowhere, fake videos that look real have started shaking up how people trust what they see online. Thanks to tools now within reach, making convincing fakes takes little time, effort, or money. Not every use causes harm - some help in movies, games, or classrooms. Yet trouble arises when someone twists them to lie, cheat, or ruin another's name. Voices and faces twisted by code can shift beliefs, sway voters, or steal cash. Because of this, spotting the fakes fast and accurately matters more than ever in today's digital world.

Something different happens here - this method uses a mix of deep learning tools to spot fake videos better. Instead of just looking at single images, it pulls together ResNeXt-50, good at catching details in pictures, along with LSTM networks, which track changes over time. One way it works is by watching how faces move across several frames, not only what they look like up close. Because of this blend, small glitches in blinking, head movement, or lighting shifts become easier to catch. Traditional methods miss those signs since they examine snapshots one after another without linking them. So while others fall short, this setup tracks both space and time clues more closely.

What makes the ResNeXt-LSTM setup stand out is how it picks up tiny details in video images. Thanks to the ResNeXt-50 part, using grouped convolutions, it detects fine textures, odd edges around faces, strange light shifts, or glitches left by fake-making tools. Spotting these image-level clues helps tell real clips apart from altered ones. After that, the LSTM takes over - looking at how those detected traits shift across consecutive frames. It catches awkward motion jumps or timing hiccups often missed when viewing stills alone. Together, reading both space-based signs and time-linked changes gives this method its edge.

Testing the new model happened with two popular video collections: Celeb-DF v1 along with DeepForensics++. Videos inside include many real clips mixed with altered ones made through various deepfake methods. Because of their wide variety, these sets act like actual conditions where detection tools need to work. Training on them showed the mix-style design handles diverse edits well while staying accurate. Results proved it keeps working even against tricky, lifelike fakes.

Trying out the fresh version used two well-known video groups - Celeb-DF v1 paired with DeepForensics++. Inside those batches are plenty of genuine recordings sitting beside modified footage created using different deepfake techniques. Wide differences across samples mimic how things really unfold when spotting tools get tested in practice. Learning from such material revealed that blending styles manages varied changes effectively without losing precision. Outcomes confirmed steady performance, especially when facing tough, realistic-looking fake videos.

Looking closer, the research shows how using time and space data helps deep learning spot fakes better than old ways that just checked single images. While past systems examined one frame at a time, they missed cleverly made fake videos where each still looked real enough. Yet once movement across moments enters the picture - like awkward eye blinks, shaky turns of the face, or odd muscle shifts - something feels off. Because it tracks these motion clues, the new mixed approach catches signs of tampering that typically slip through. What slips by in isolation stands out over time.

Even though the outcomes look good, using these mixed designs brings some heavy computing demands. Because the system uses a complex ResNeXt-50 structure along with step-by-step LSTM analysis, it needs powerful hardware when learning. Looking at videos means going through many images one after another, filling up memory fast while slowing things down. That slowdown means without strong graphics processors, working with so much data would take far too long. While lighter models work faster on individual images, the new design takes more time to train and needs stronger hardware. Still, the boost in precision makes up for those demands. Better performance matters most when spotting fake videos or checking digital evidence. That trade-off fits best with tasks done after recording - like verifying files or reviewing footage. Even if it runs slower, catching subtle edits beats instant results every time.

Even so, getting detection tools to run instantly is still a key target ahead. Since today's messaging apps move fast, fake videos often go viral well before checks catch up. Because of that, coming studies need to speed up combined methods without losing precision. Ways like shrinking model size, trimming unnecessary connections, or transferring learned patterns could lighten heavy neural networks - yet keep results reliable.

One path worth exploring further involves using different types of data together to catch fake videos. Most current methods look only at images taken from moving pictures. Yet linking sound, voice traits, body signals, or written words might make spotting fakes far more accurate. Think mismatched mouth motion and spoken sounds, odd blink rates, even tiny skin shade shifts hinting at false pulse readings - all these can point toward tampering. When layered, such clues may build stronger defenses against cleverly altered media. Systems that pull in varied inputs tend to stand up better under pressure from advanced forgeries.

Understanding how AI makes choices matters more as models grow complicated. Another path forward comes from explainable artificial intelligence, where clarity shapes better deepfake spotting tools. Instead of guessing why a result appears, users see which parts of a video frame influenced the outcome. Visual cues reveal what the system focused on during analysis. This openness builds confidence while helping experts dig into findings with precision. Researchers gain insight not just into results but into reasoning behind them. When detection methods show their workings, alignment with rules and responsible practices grows stronger. Clarity becomes part of the design rather than an afterthought. Tools like these shift expectations about what trustworthy tech should do.

Beyond just showing how they work, newer designs like Vision Transformers open fresh paths to better spotting fakes. Because these models handle distant image details so well, they've done impressively across many vision jobs. When paired with older convolution methods, transformers might build stronger tools for studying tricky visual changes in altered content. Lighter mix-models - linking deep nets with simpler learning rules - could run smoothly even on low-power gadgets.

Figuring out how well deepfake detection works across different scenarios is a big deal. Since tools that create fake content keep changing, fresh tricks pop up all the time. Models taught on old data might fail when faced with something they have never seen before. One way around this? Scientists are testing methods like learning without labels, adjusting between environments, or training how to learn faster. The idea behind these paths lies in building systems that adapt - catching fakes even if they look unlike anything known.

Right now, how we handle ethics shapes what comes next for deepfake detection tools. Though these systems can reduce damage from fake videos, care is needed so they are not twisted into something dangerous. Trust sticks around only if the tech acts fairly, shows its workings clearly, and answers for mistakes. Oddly enough, progress depends less on code and more on teamwork - scientists, firms, lawmakers, agencies all leaning in together. Only through that mix can solid rules grow, guiding how such powerful tools should behave.

Surprisingly, this work shows how mixing deep learning methods that handle space and time can boost detection of fake videos. Because it links ResNeXt50 with LSTM, the system picks up strange image details along with odd movements in altered footage. From tests on Celeb-DF v1 and DeepForensics++, solid numbers back its strength - accuracy stays strong, just like balance among scores. Even when heavy computing loads pop up, such combined designs still push forward better trust and durability in spotting fakes.

Deepfakes keep changing, so spotting them must change too. Because these fakes get smarter, tools to catch them cannot stay behind. Progress comes not just from code but from teams across fields sharing ideas. When researchers mix ethics with smart algorithms, results start holding up under pressure. Trust online depends on systems that see through deception before it spreads. Machines learn fast - so should our ways of checking what is real

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APPENDIX 1:

Research Paper: SachAI: The DeepFake Video Detection Model

SachAI: The Deep Fake Video Detection Model

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Abstract—These rapid advancements in deepfake technology have made it easier and cheaper to create artificial lifelike films and images, and have significantly improved the security, privacy and confidentiality of digital media. The detection of such of these alterations is an important research topic because the existing forensic techniques often do not generalize them across the different datasets and modification methodologies. In this paper, we propose a deep learning-based method or approach for excellent deepfake video detection. To capture the temporal dependencies between successive video frames, ResNeXt's representational power for spatial feature extraction is combined with Long Short-Term Memory (LSTM) networks. The proposed method combines the spatial and temporal analysis to improve the ability to identify small variations such as facial expressions and motion patterns, which are often and hence overlooked in the standalone frame-level identification. Also addition to the expanding corpus of research targeted at enhancing the dependability of deepfake detection systems.

Index Terms—Deepfake detection, ResNeXt, LSTM, video forensics, deep learning, spatiotemporal analysis

I. INTRODUCTION

The phrase "deepfake," which usually refers to the synthetic media where a person's voice, likeness, or behaviors are digitally altered and changed intentionally to produce a hyper-realistic yet fake content, which is derived from the words "deep learning" and "fake." Deepfake technology has advanced quickly and rapidly since its first prominent appearance in 2017, enabling even some non-experts to create and modify videos with easily and operationally accessible tools. Deepfakes then present significant risks, such as the circulation of false information, identity theft, political manipulation, and invasions of privacy, even while they have promise uses in accessibility, education, and entertainment. While differentiating these phoney movies from the real content has becoming increasingly difficult in cybersecurity and the digital forensics due to their increasing growing sophistication. [1]–[4]. Sophisticated deepfake detection methods have been developed in some recent years. In the face of increasingly intricate manipulations, conventional methods like manually constructed feature extraction and simple machine learning

have proven insufficient. Therefore, scientists are now employing deep learning-based methods that leverage the temporal and spatial features of movies. Studies have looked into using convolutional neural networks (CNNs), hybrid architectures including CNN-LSTM models, and ensemble approaches that blend 2D and 3D CNNs to record spatiotemporal differences. Other studies have also examined frequency-domain features, hybrid CNN-based frameworks, adversarial robustness, and novel architectures such as CNN-LSTM-Transformer models. These highlight the current struggle between the deepfake technology generation and those forensic detection methods. [5]–[8].

II. LITERATURE REVIEW

A. Datasets, Challenges, and Research Gaps

The availability of datasets has significantly affected research in deepfake detection. The Face Forensics++ dataset was used along with Celeb-DF dataset. Various films of celebrities with challenging facial alterations, despite the small scale. The serious difficulty of generalization in the actual world is demonstrated by the fact that even the top-performing models in the DeepFake Detection Challenge (DFDC, Facebook 2019) only achieved on the hidden test set. The challenge tried to address diversity and size. [9]–[12]. Models frequently learned dataset-specific, generalization—which was noted by Verdoliva (2020) and Yu et al.(2021)remains one of the most urgent problem. Despite evidence from Xu et al. (2022) and Güera and Delp (2018) that motion artefacts which are useful indications of the fraud, and temporal modelling has also been underused. From their susceptibility to simulation. The computational expense of deep models, which limits real-time deployment in forensic and security applications, was also highlighted by Kaur et al. (2024). Moreover, Xu et al. (2022) warned that the generative models are rapidly developing. However this multimodal detection which usually relies on the multiple content modalities, which includes the lip-sync discrepancies is normally used to detect the manipulated content (Masood et al. 2023) [13]–[15].

TABLE 1
SUMMARY OF DEEPPAKE DETECTION LITERATURE

Author(s), Year	Methodology / Contribution	Dataset(s) Used	Key Findings	Limitations
Güera and Delp (2018)	CNN for frame-level visual features + RNN for temporal dependencies	UADFV	Temporal irregularities (lip-sync jitter, uneven transitions) were detected	Moderate accuracy, requires larger datasets [7], [8]
Afchar (2018)	MesoNet (shallow CNN for Nanoscale features)	UADFV	Lightweight, ~83% accuracy	Struggled with high-quality manipulations [2]
Rossler (2019)	XceptionNet benchmarks	FaceForensics++	Deeper architectures outperformed shallow ones, achieving approx. 81% accuracy on compressed videos	Controlled dataset conditions [3]
Li et al. (2020)	Face X-ray (blending boundary detection)	Celeb-DF, FaceForensics++	New forensic clues from pixel-level boundary detection	Dependent on distinct boundary artifacts, less effective on newer models [4]
Verdoliva (2020)	Compared deep learning vs. traditional forensic methods	Multiple	Concluded deep learning is superior in versatility	Vulnerable to adversarial attacks, overfitting [5]
Rana et al. (2022)	Classified methods into ML, statistical, blockchain, and DL	Multiple	Broader taxonomy beyond deep learning	Scalability issues [8]
Xu et al. (2022)	Review of 300+ papers	Multiple	Declared the detection-generation process to be an "arms race"	Highlighted persistent limitations in detectors [9]
Masood et al. (2023)	Multimodal detection (audio + video)	Custom	Improved detection of lip-sync	Vulnerable to advanced mimicry [10]
Patil (2023)	Biological cues (pulse, blink frequency)	Custom	New characteristics for detection	Generators increasingly simulate bio-signals [11]
Gambín et al. (2024)	Blockchain-based provenance verification	N/A	Ensures authenticity tracking	Scalability and deployment challenges [12]
Gong and Li (2024)	Comparative analysis of transformers, CNNs, semi-supervised, and bio-signal techniques	Multiple	Demonstrated use of transformers in forensics	Computational cost [13]
Kaur et al. (2024)	Comprehensive review	Multiple	Issues highlighted include weak generalizability, high computing cost, and dataset imbalance	Open challenges remain [14]



Fig. 1. Flow Chart

III. METHODOLOGY

Deepfake videos are you know very difficult to detect due to their extraction of the spatial and temporal feature, which is kind of difficult to distinguish the manipulated videos apart from the real ones. This article presents four methods mainly dataset creation, data augmentation, ResNeXt-50-based feature extraction and LSTM-based temporal feature extraction model.

A. Dataset

Two reference datasets, Celeb-DF v1 and DeepForensics++ were mainly used for the experimental validation. Celeb-DF v1 is very difficult as a dataset to generalize because it includes videos of celebrities in a very diverse environments, positions, and with varied facial expressions. DeepForensics++ incorporates large-scale altered videos, created using various manipulation techniques such as face swapping and recreation to provide a comprehensive set of real and fake samples for the training and testing. These datasets additionally ensure a thorough evaluation of the proposed model using various manipulation methods.

B. Pre-processing

The raw video data was pre-processed to remove the unnecessary frames at the predefined in-order of interval in order to conserve or preserve the temporal coherence. Moreover, each frame was normalized and then compressed to a straight standard of resolution in order to accelerate the training convergence. The facial features were highlighted by cropping and face alignment using popular detection libraries. The pixels

cutting method reduces computer complexity while maintaining sufficient temporal information for sequential analysis.

C. Model Architecture

The detection system is a hybrid model that blends with easily ResNeXt-50 and LSTM. ResNeXt-50, a ResNet variation that employs grouped convolutions for the effective representation learning which has been used to extract the fine-grained spatial data from the individual video frames. These characteristics record a minute manipulation-induced artifacts in texture, lighting, and face borders. The extracted frame-level features were then further sequentially fed into an LSTM network to describe temporal relationships and thus the motion inconsistencies over those succeeding frames.

D. Training Setup

The model was trained by supervised learning using balanced sets of an actual and fake videos. During training over multiple epochs and those mini-batch processing was employed to reduce the overfitting and stabilize learning. Metrics including precision, recall and a few other variables were then used to evaluate the framework's performance.

IV. RESULTS

The Celeb-DF v1 and DeepForensics++ datasets were then used to assess suggested deepfake detection framework and using a standard performance metrics which is such as accuracy, precision, recall and F1-score. These metrics are then finally summarized in the Table 1. II.

TABLE II
PERFORMANCE METRICS OF PROPOSED MODEL

Accuracy	Precision	Recall	F1-Score
97%	92.6%	94.3%	93.5%

A confusion matrix was afterwards created in order to get a better understanding of our model categorization performance. The system showed a high capacity to differentiate between a real and deepfake samples correctly detecting the majority of these cases with only a few incorrect classifications. The ResNeXt-50 and LSTM hybrid model can detect minute spatiotemporal anomalies in motion patterns and face features that are often overlooked by frame-level detection algorithms as demonstrated by this exceptional balance between recall and precision. Our results demonstrate the dependability of the proposed approach which is further validated using a k-fold cross-validation to ensure generalization and then lower the risk of overfitting. By the utilization of manipulation-agnostic spatiotemporal signals the suggested ResNeXt-LSTM model enhances the generalization and also performs well on the two difficult datasets. The model is mostly ideal for offline or serve type of forensic applications with the clear paths for optimization in future work for sure, even though real-time deployment on low resource devices has been constrained by the computing demands.

V. CASE STUDIES

Deepfakes are a real world menace which extends far beyond the novelty and amusement. Synthetic media is nowadays increasingly being used as a weapon in critical industries including politics, banking and governance where even a single phony audio or video clip can result in widespread disinformation, reputational damage or significant fraud. The two case studies that follow illustrate different but equally urgent issues raised by deepfakes and show how cutting-edge detection tools—specifically, those created by Reality Defender—have been used to mitigate these threats in real-world scenarios [16], [16].

A. Political Disinformation in Taiwan

The influence of deepfakes on political events became evident during Taiwan's 2024 presidential election. A disinformation campaign targeted Ko Wen-je, the Taiwan People's Party (TPP) candidate and former mayor of Taipei, using a fabricated audio recording. The manipulated clip imitated Ko's voice, making derogatory remarks about his rival, Lai Ching-Te of the Democratic Progressive Party (DPP), and included false claims of voter manipulation and foreign interference. The success of the attack centered on two things: first one was, Ko's long public speeches which were easily available, hence offering enough information to train a voice clone with easily accessible, cheap tools; and second, the manipulated content was rapidly shared by the Chinese state media, thus reaching a much wider audience before any verification could be done. And then at last the voters were confused and too suspicious in every way. [17].

This case emphasizes on how voting has the integrity which is vulnerable in the digital age since there misinformation can spread faster than later corrections. It also demonstrates that how deepfakes can undermine the political institutions which alter the public opinion and have an impact on political results. In addition it also highlights the vital need for the quick identification which improves the transparency and accountability in governance and thus lessens its reputational damage.

B. Preventing Deepfaked Bank Fraud

While political institutions faces significant challenges from the disinformation and now unfortunately financial systems are also at a huge risk from the deepfake-based fraud. A major international bank which handle the trillions of dollars worth of assets and billions of transactions every day suddenly saw an increase in the fraud phone calls. Cybercriminals or hackers or phishers in some case also used to attempted to evade the regular verification procedures and thus enable unlawful fund transfers by impersonating the clients and workers using a heavy deepfaked voice. [18].

By using a particular and special strategy the bank was able to successfully detect and combat/fight the fraudulent attempts thus protecting its clients and institutional assets. Importantly, the system was designed to function in a range of scenarios, accounting for different accents. the adaptability of

this detection technique demonstrates its application outside of banking, including verifying customer service letters, screening interview tapes, and monitoring social media platforms for fraudulent audio content.

VI. CHALLENGES AND OPPORTUNITIES

A. Challenges

Several challenges persist in deepfake detection:

- 1) **Rapid Evolution of Generation Techniques:** The techniques used to create deepfakes are developing quickly, creating increasingly real fakes that very much resemble real videos. As a result, detection models often lag behind, becoming less effective against newly emerging manipulation techniques. [19].
- 2) **Variability in Video Quality:** The performance of detection systems can be heavily influenced by factors such as resolution, compression, noise, and the platform on which the video is shared. A model trained on high-quality datasets may fail when confronted with low-resolution or highly compressed videos common in real-world social media platforms .
- 3) **Adversarial Attacks:** Detection systems themselves are susceptible to an adversarial manipulation. There is an urgent need for robust and durable detection techniques since attackers can subtly alter forged films to trick the detectors. [20].
- 4) **Lack of Standardized Benchmarks:** Large and excellent labeled datasets are nowadays necessary for a deep learning-based detection technique. However it is expensive and very time-consuming to gather and annotate those datasets. Which Ensures that data accuracy is challenging and problematic for many businesses which has a direct effect on detection performance. [21].

B. Opportunities

Deepfake technology's explosive growth in 2025 has produced a dynamic environment for detection method research and innovation, with a number of high-impact prospects showing up in the literature. An Accuracy above 95compression levels and manipulation techniques have been demonstrated by hybrid deep learning approaches that integrate spatial and temporal analysis, such as RLNet, which combines LSTM for temporal sequence modelling with ResNet for frame-level feature extraction. In order to counteract live-streamed manipulations and false information on social media, detection research is also shifting toward multimodal and real-time solutions, combining visual, physiological (such as heart rate and eye blinking), aural, and verbal clues.Explainable AI (XAI) is another important area [22], [23]. Frameworks such as DF-P2E combine natural language explanations and Grad-CAM saliency maps to improve transparency, foster user confidence, and assist with the regulatory compliance. The problems of generalization, adversarial robustness, and dataset diversity still exist in spite of these developments, which motivate research into a self-supervised learning, meta-learning, domain adaptability, and adversarial training to withstand

changing generative models. However, in order to guarantee equitable benchmarking and prevent abuse, experts stress the significance of ethical design, standardization, and cooperative dataset production. The potential of Vision Transformers (ViTs) as CNN substitutes, lightweight hybrid systems with XGBoost for deployment on devices with limited resources, and the combination of behavioural context analysis and emotion analysis with video forensics to enhance authenticity assessment are further highlighted by emerging research directions. [24].

VII. CONCLUSION

With increasingly realistic manipulations endangering privacy, politics, finance, and social stability, the spread of deepfake technology presents a serious threat to digital trust. In order to solve the issue, this study suggested a hybrid deep learning framework that combines Long Short-Term Memory (LSTM) networks for temporal sequence modelling with ResNeXt-50 for spatial feature extraction. By combining the motion dynamics with frame-level extraction information, the model performs well on the benchmark datasets such as Celeb-DF v1 and DeepForensics++, with the accuracy of 97% demonstrate how successfully spatiotemporal integration detects subtle abnormalities that basically the previous traditional frame based approaches ignore or could not operate on. The deep ResNeXt-50 backbone and sequential LSTM processing in the ResNeXt-LSTM architecture result in a moderate computational overhead. The model needs GPU acceleration during training so that it can easily interpret the video sequences effectively and especially as the temporal coherence is normally preserved by analyzing very frames per video. Therefore, as a result, training takes longer than lightweight CNN-only methods particularly when dealing with huge datasets. However novel strategies like explainable AI or multimodal fusion or even infact some of these lightweight models also transformer-based systems frequently present encouraging avenues for further future study. The study stresses how hybrid architectures boost the accuracy and dependability of deepfake detection when utilized with temporal and geographical information. However, if we consider it, the problem of fake media will always exist. In order to create scalable and accurate real-time detection systems that safeguard digital authenticity, further advancement necessitates technical innovation, moral principles, and joint research.

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